Week 12: WTP & Simulations

November 11, 2026

THE POINT OF THIS WEEK

A table of β coefficients is hard to act on. "Electric powertrain: −4.0" means nothing to a product manager. This week turns the estimates into the two outputs a design decision actually needs:

1. **Willingness to pay (WTP)**: what each attribute is worth, *in dollars*.
2. **Market simulations**: the share of the market a product would capture against specific competitors.

Both are arithmetic on the coefficients you already estimated. Both also inherit the coefficients' uncertainty, so both get confidence intervals the same way: take draws of the coefficients and push every draw through the calculation.

WILLINGNESS TO PAY

Split utility into price and everything else:

$$\tilde{u}_j = \alpha p_j + \boldsymbol{\beta} x_j + \tilde{\varepsilon}_j$$

α is the utility of one more unit of price (negative), and each β is the utility of one more unit of an attribute. Dividing one by the other cancels the utility units and leaves dollars:

$$\boldsymbol{\omega} = \frac{\boldsymbol{\beta}}{(-\alpha)}$$

WTP comes out **in the units price was coded in**. If price is in $1,000s, a WTP of 0.14 means $140.

Hats mean "estimated." α and β are the true (unknown) parameters; $\hat{\alpha}$ and $\hat{\beta}$ are what your model returned, so what you compute is $\hat{\omega} = \hat{\beta}/(-\hat{\alpha})$.

Worked example

Cars, with price in $1,000s, fuel economy in mpg, and an electric dummy. Estimates: $\hat{\alpha} = -0.7$, $\hat{\beta}_1 = 0.1$, $\hat{\beta}_2 = -4.0$.

Attribute	WTP	In dollars
Fuel economy	0.1 / 0.7 = 0.143	+$143 per mpg
Electric	−4.0 / 0.7 = −5.71	−$5,714 vs. gas

```
beta <- c(price = -0.7, mpg = 0.1, elec = -4.0)
wtp <- beta / (-1 * beta['price'])
```

The price entry divides to −1 by construction; ignore it.

WTP WITH UNCERTAINTY

A ratio of two uncertain numbers has no simple standard error formula, so **simulate it.** Same recipe as the uncertainty week:

1. Get the coefficients and the covariance matrix (the negative inverse of the hessian).
2. Draw many coefficient vectors from a multivariate normal.
3. Compute WTP for every draw.
4. Report the mean and the 2.5% and 97.5% quantiles.

```
hessian <- matrix(c(
  -6000,   50,   60,
     50, -700,   50,
     60,   50, -300),
  ncol = 3, byrow = TRUE)
covariance <- -1 * solve(hessian)

draws <- MASS::mvrnorm(10^4, beta, covariance)
wtp_draws <- draws[, 2:3] / (-1 * draws[, 1])

logitr::ci(wtp_draws)   # mean, lower, upper
```

From an estimated {logitr} model, `coef(model)` and `vcov(model)` give you the two inputs directly (`code/6.1-compute-wtp.R` in `logitr-cars`):

```
coefs <- coef(model_mnl)
covariance <- vcov(model_mnl)
coef_draws <- as.data.frame(
  MASS::mvrnorm(10^4, coefs, covariance))
wtp_draws <- -1 * (coef_draws[, 2:5] /
                   coef_draws[, 1])
ci(wtp_draws, level = 0.95)
```

ESTIMATING WTP DIRECTLY

Instead of computing WTP after the fact, rewrite the model so the WTP values *are* the parameters. Substitute $\boldsymbol{\omega} = \boldsymbol{\beta}/(-\alpha)$ and define the scale $\lambda = -\alpha$:

$$\tilde{u}_j = \lambda(\boldsymbol{\omega} x_j - p_j) + \tilde{\varepsilon}_j$$

Space	Estimates	WTP	Log-likelihood
Preference	$\alpha, \boldsymbol{\beta}$	Computed	Convex
WTP	$\lambda, \boldsymbol{\omega}$	Estimated	**Non-convex**

A non-convex log-likelihood can have several local maxima, and the optimizer stops at whichever one is closest to where it started. The fix is a **multi-start**: run the optimization from many random starting points and keep the best result.

```
model_mnl_wtp <- logitr(
  data    = data,
  outcome = "choice",
  obsID   = "obsID",
  pars = c('fuelEconomy', 'accelTime',
           'powertrain_Electric',
           'powertrain_Hybrid'),
  scalePar = 'price',     # the WTP-space switch
  numMultiStarts = 10)
```

Price moves out of `pars` and into `scalePar`. Then check that you found a real maximum, not just a stopping point:

```
# 1st order: gradient should be ~0
model_mnl_wtp$gradient

# 2nd order: all eigenvalues negative
eigen(model_mnl_wtp$hessian)$values

# Same WTP and log-likelihood as the
# preference-space model?
wtpCompare(model_mnl, model_mnl_wtp,
           scalePar = 'price')
```

If the WTP-space log-likelihood is noticeably worse than the preference-space one, the multi-start did not find the global maximum. Add more starts.

SIMULATING MARKET SHARES

A **market** is a set of alternatives a buyer chooses among: your product plus the competitors you expect it to face. Write it as a matrix $\boldsymbol{X}$, one row per alternative, one column per attribute. Then apply the logit formula with your estimates:

$$\hat{P}_j = \frac{e^{\hat{\beta}' \boldsymbol{x}_j}}{\sum_{k=1}^{J} e^{\hat{\beta}' \boldsymbol{x}_k}}$$

In R, `X %*% beta` computes every alternative's observed utility at once.

Worked example

Same coefficients as above, with β ordered as price, mpg, electric:

Alt	Price	mpg	Elec	v_j	Share
1	15	20	0	−8.5	81.6%
2	30	100	1	−15.0	0.1%
3	20	40	0	−10.0	18.2%

```
X <- matrix(c(
  15,  20, 0,
  30, 100, 1,
  20,  40, 0),
  ncol = 3, byrow = TRUE)
```

```
v_j <- X %*% beta
P_j <- exp(v_j) / sum(exp(v_j))
```

Shares depend on **differences** in utility, so a 6.5-point utility gap leaves the electric car with almost nothing.

With uncertainty: `predict()`

`predict()` does the draws for you: it samples coefficients, computes the shares for each draw, and returns the mean and the confidence interval (`code/7.1-compute-market-sims.R`):

```
baseline <- data.frame(
  altID = c(1, 2, 3),
  obsID = c(1, 1, 1),
  price = c(15, 25, 21),
  fuelEconomy = c(20, 100, 40),
  accelTime = c(8, 6, 7),
  powertrain_Electric = c(0, 1, 0),
  powertrain_Hybrid   = c(0, 0, 1))

sim_mnl <- predict(
  model_mnl,
  newdata    = baseline,
  obsID      = 'obsID',
  level      = 0.95,
  interval   = 'confidence',
  returnData = TRUE)
```

The column names in `newdata` must match the model's `pars` exactly, dummy coding included. To run several markets at once, stack them in one data frame with a different `obsID` for each scenario.

> WTP and market shares are both functions of uncertain coefficients, so they are uncertain too. Never report a WTP or a share as a single number: report the mean with a 95% confidence interval from the coefficient draws.

PRESENTING THE RESULTS

WTP

- **Scale WTP to a change people care about.** WTP is per unit, and "$143 per mpg" undersells a 10 mpg improvement. Multiply the mean *and* both bounds by the size of the change.
- **Flip the sign for "lower is better" attributes.** A 3-second faster 0–60 time is a change of −3, so multiply by −3.
- **Drop λ before plotting** a WTP-space model's draws. It is the price scale, not a WTP.
- **Continuous attributes** plot as a line with a ribbon; **categorical** ones as points with error bars, with the reference level at 0. Put every panel on the same y-axis so the sizes compare honestly.

```
wtp_ci <- ci(wtp_draws, level = 0.95)
wtp_ci <- wtp_ci[-1, ]   # drop lambda
```

Market shares

A bar per alternative with an error bar for the interval, y-axis fixed from 0 to 100%. One panel per scenario with `facet_wrap(~obsID)`.

```
sim_mnl %>%
  ggplot(aes(
    x = label,
    y = predicted_prob,
    ymin = predicted_prob_lower,
    ymax = predicted_prob_upper)) +
  geom_col(fill = "grey", width = 0.6) +
  geom_errorbar(width = 0.3) +
  scale_y_continuous(
    limits = c(0, 1),
    labels = scales::percent)
```

Common mistakes

Symptom	Usual cause
WTP has the wrong sign	Divided by α instead of $-\alpha$
WTP off by 1,000×	Forgot price was coded in $1,000s
WTP-space LL worse than pref-space	Too few multi-starts
`predict()` errors on `newdata`	Column names don't match `pars`
Shares don't sum to 1	Alternatives from different markets share an `obsID`
Huge WTP intervals	Price coefficient is close to zero

BEFORE NEXT CLASS

- Final Survey — due **Nov 15**.
- HW 11: Heterogeneity — due **Nov 16**.
- With your team, compute WTP from your pilot model, re-estimate it in WTP space, and set up one or two market scenarios pitting your product against real competitors. All of it goes into the Final Analysis report, due **Dec 6**.
- Quiz 5 is at the start of class on **Nov 18**.

You should be able to:

- Compute WTP from preference-space coefficients and state it in dollars.
- Get a 95% confidence interval on WTP by simulating coefficient draws.
- Estimate a WTP-space model with `scalePar` and a multi-start, and check its first- and second-order conditions.
- Explain why WTP-space models need a multi-start.
- Compute market shares for a set of alternatives by hand and with `predict()`, with confidence intervals.